

Project Title: Recognising everyday activities from lower back and thigh-mounted accelerometers in free-living conditions

Question: Can a machine learning model trained on some participants' lower back and thigh-mounted accelerometers correctly classify the activities of unseen participants?

Dataset: I will use the HARTH (Human Activity Recognition Trondheim) dataset, from the UCI Machine Learning Repository (<https://doi.org/10.24432/C5NC90>), released under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence.

Citation for journal paper: A. Logacjov, K. Bach, A. Kongsvold, H. Bårdstu, and P. Mork, "HARTH: A Human Activity Recognition Dataset for Machine Learning," *Sensors*, vol. 21, no. 23, p. 7853, Nov. 2021, doi: 10.3390/s21237853.

Citation for raw dataset: A. Logacjov, A. Kongsvold, K. Bach, H. Bårdstu, and P. Mork. "HARTH," UCI Machine Learning Repository, 2021 [Online]. Available: <https://doi.org/10.24432/C5NC90>.

Target variable and features: The target variable is the activity label (reduced from 12 to 8 classes: sitting, walking, standing, lying, running, shuffling, stairs and cycling). Ascending and descending stairs will be merged, as will sitting and standing cycling. The two inactive cycling labels will be dropped because they have less than 15 minutes of data and are likely to resemble sitting and standing. The main input features are the three axes of each accelerometer and each sensor's vector magnitude (calculated as $\sqrt{x^2 + y^2 + z^2}$). Data will be segmented into non-overlapping one second windows as in Logacjov et al. (2021), and summarised by the mean, standard deviation and range.

Records and data-quality: The dataset contains 22 participants, and approximately 35 hours, giving about 125000 one-second windows. Initial data inspection found differences in sampling frequency (one participant at 100 Hz, to be resampled to 50 Hz), extra columns, small timestamp gaps, strong class imbalances, and some activities coming mostly from few a few participants so the train/test split needs to be chosen carefully.

Proposed baseline, candidate models and evaluation metrics: The proposed baseline always predicts the most common class (sitting). Candidate models are a decision tree, k-nearest neighbours (classifies by similarity to training windows) and Support Vector Machines (SVM) (efficient for multiclass classification when boundaries are clear). A few participants will be held out as a test set, chosen so every class is present and models will be tuned using cross-validation grouped by participants on the rest. This prevents leakage, since a model tested on people it trained on could learn to recognise the person rather than the activity. The main metric is the F1 score since it weighs classes evenly despite imbalances, supported by a confusion matrix showing which activities are confused with each other.

Responsible-ML: The data is anonymised and publicly licensed, and raw files aren't stored in the repo. The participants included 22 healthy adults aged between 25 and 68 so the model might not work well children, older adults or individuals with conditions that affect how they move. Labels were annotated manually from video, so some may be wrong, especially at transitions between activities. A model like this could be misused for workplace monitoring without consent, so it is intended for research and education only, not for monitoring individuals.

Repository: <https://github.com/sarabartolo/harth-free-living-activity-recognition.git>